

● MEDIUM

● ALGEBRA

● ANSWER KEY

Linear equations in one variable

02

10 answers · Rationale-rich · Teach from doc

Q1**B****B.** $4w + 12$

Choice B is correct. It is given that the width of the dance floor is w feet. The length is 6 feet longer than the width; therefore, the length of the dance floor is $w + 6$. So the perimeter is $w + w + (w + 6) + (w + 6) = 4w + 12$.

Choice A is incorrect because it is the sum of one length and one width, which is only half the perimeter. Choice C is incorrect and may result from using the formula for the area instead of the formula for the perimeter and making a calculation error. Choice D is incorrect because this is the area, not the perimeter, of the dance floor.

Q2**C****C.** 29

Choice C is correct. Subtracting $4x + 4$ from both sides of the given equation yields $1x + 4 = 29$, or $x + 4 = 29$. Therefore, the value of $x + 4$ is 29.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of x , not $x + 4$.

Choice D is incorrect and may result from conceptual or calculation errors.

Q3**B****B.** $a = 2$ and $b = 7$

Choice B is correct. Distributing the a on the left-hand side of the equation gives $3a - b - ax = -1 - 2x$. Rearranging the terms in each side of the equation yields $-ax + 3a - b = -2x - 1$. Since the equation has infinitely many solutions, it follows that the coefficients of x and the free terms on both sides must be equal. That is, $-a = -2$, or $a = 2$, and $3a - b = -1$. Substituting 2 for a in the equation $3a - b = -1$ gives $3(2) - b = -1$, so $b = 7$.

Choice A is incorrect and may be the result of a conceptual error when finding the value of b . Choices C and D are incorrect and may result from making a sign error when simplifying.

Q4**C****C.** $950 + 50(t - 2) = 1,100$

Choice C is correct. It's given that the cost to rent a commercial fishing boat is \$ 950 for the first 2 hours and an additional \$ 50 per hour for each hour after the first 2 hours. It's also given that t represents the total number of hours and $t > 2$. Therefore, the number of additional hours after the first 2 hours can be represented with the expression $t - 2$. The cost for these additional hours is \$ 50 per hour, so the cost for the additional hours can be represented by the expression $50t - 2$. The total cost can be calculated by adding the cost for the first 2 hours to the cost for the additional hours and can be represented by the expression $950 + 50t - 2$. It's also given that the total cost to rent the commercial fishing boat from the company for t hours is \$ 1,100. Thus, the equation that represents this situation is $950 + 50t - 2 = 1,100$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Q5**34**

The correct answer is 34. It's given that a line segment has a length of 115 cm and is divided into three parts, where one part is 47 cm long and the other two parts have lengths that are equal. If x represents the length, in cm, of each of the two parts of equal length, then the equation $47 + x + x = 115$, or $47 + 2x = 115$, represents this situation. Subtracting 47 from each side of this equation yields $2x = 68$. Dividing each side of this equation by 2 yields $x = 34$. Therefore, the length, in cm, of one of the two parts of equal length is 34.

Q6**B**

B. $\frac{17}{2}$

Choice B is correct. Because 2 is a factor of both $2x$ and 6, the expression $2x - 6$ can be rewritten as $2(x - 3)$. Substituting $2(x - 3)$ for $(2x - 6)$ on the left-hand side of the given equation yields $3(2)(x - 3) - 11 = 4(x - 3) + 6$, or $6(x - 3) - 11 = 4(x - 3) + 6$. Subtracting $4(x - 3)$ from both sides of this equation yields $2(x - 3) - 11 = 6$. Adding 11 to both sides of this equation yields $2(x - 3) = 17$. Dividing both sides of this equation by 2 yields $x - 3 = \frac{17}{2}$.

Alternate approach: Distributing 3 to the quantity $(2x - 6)$ on the left-hand side of the given equation and distributing 4 to the quantity $(x - 3)$ on the right-hand side yields $6x - 18 - 11 = 4x - 12 + 6$, or $6x - 29 = 4x - 6$. Subtracting $4x$ from both sides of this equation yields $2x - 29 = -6$. Adding 29 to both sides of this equation yields $2x = 23$.

Dividing both sides of this equation by 2 yields $x = \frac{23}{2}$. Therefore, the value of $x - 3$ is $\frac{23}{2} - 3$, or $\frac{17}{2}$.

Choice A is incorrect. This is the value of x , not $x - 3$. Choices C and D are incorrect. If the value of $x - 3$ is $\frac{15}{2}$ or $-\frac{15}{2}$, it follows that the value of x is $\frac{21}{2}$ or $-\frac{9}{2}$, respectively.

However, solving the given equation for x yields $x = \frac{23}{2}$. Therefore, the value of $x - 3$ can't be $\frac{15}{2}$ or $-\frac{15}{2}$.

Q7**D****D. 44**

Choice D is correct. It's given that the candle starts with 17 ounces of wax and has 6 ounces of wax remaining after a period of time has passed. The amount of wax the candle has lost during the time period can be found by subtracting the remaining amount of wax from the amount of wax the candle was made of, which yields $17 - 6$ ounces, or 11 ounces. This means the candle loses 11 ounces of wax during that period of time. It's given that the amount of wax decreases by 1 ounce every 4 hours. If h represents the number of hours the candle has been burning, it follows that $\frac{1}{4} = \frac{11}{h}$. Multiplying both sides of this equation by $4h$ yields $h = 44$. Therefore, the candle has been burning for 44 hours.

Choice A is incorrect and may result from using the equation $\frac{1}{4} = \frac{h}{11}$ rather than $\frac{1}{4} = \frac{11}{h}$ to represent the situation, and then rounding to the nearest whole number.

Choice B is incorrect. This is the amount of wax, in ounces, remaining in the candle, not the number of hours it has been burning.

Choice C is incorrect and may result from using the equation $\frac{1}{4} = \frac{6}{h}$ rather than $\frac{1}{4} = \frac{11}{h}$ to represent the situation.

Q8**A****A. The monthly sales revenue, in dollars, from selling x tape dispensers**

Choice A is correct. It's given that the equation $15,000 = 2.00x - 4,500$ represents this situation for a month where x tape dispensers are created and sold. It's also given that the company calculates its monthly profit, in dollars, by subtracting its fixed monthly costs, in dollars, from its monthly sales revenue, in dollars. It follows that $2.00x$ represents the monthly sales revenue, in dollars. Therefore, the best interpretation of $2.00x$ in this context is the monthly sales revenue from selling x tape dispensers.

Choice B is incorrect. This is the best interpretation of 2.00, not $2.00x$.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect. This is the best interpretation of 4,500, not $2.00x$.

Q9**B**

B. The average number of feet that the tree grew per year

Choice B is correct. The height of the tree at a given time is equal to its height when it was planted plus the number of feet that the tree grew. In the given equation, 14 represents the height of the tree at the given time, and 6 represents the height of the tree when it was planted. It follows that $2n$ represents the number of feet the tree grew from the time it was planted until the time it reached a height of 14 feet. Since n represents the number of years between the given time and the time the tree was planted, 2 must represent the average number of feet the tree grew each year.

Choice A is incorrect and may result from interpreting the coefficient 2 as doubling instead of as increasing by 2 each year. Choice C is incorrect. The height of the tree when it was 1 year old was $2(1)+6=8$ feet, not 2 feet. Choice D is incorrect. No information is given to connect the growth of one particular tree to the growth of similar trees.

Q10**42**

The correct answer is 42. The expression $x + 6$ is a factor of both terms on the left-hand side of the given equation. Therefore, the given equation can be written as $x + 6\frac{1}{3} - \frac{1}{2} = -8$, or $x + 6\frac{1}{6} = -8$. Multiplying each side of this equation by -6 yields $x + 6 = 48$. Subtracting 6 from each side of this equation yields $x = 42$. Therefore, the value of x that is the solution to the given equation is 42.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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